Measurement-Process-Aware Causal Trajectory Modeling for Transportable Early Sepsis Prediction: A Multi-Center Retrospective Evaluation

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***Abstract*—Machine-learning early-warning models for sepsis in intensive care units (ICUs) often lose accuracy when deployed at a different institution, in part because they exploit site-specific monitoring and laboratory-ordering practices as predictive shortcuts. We present AegisSepsis AI, an observation-process-aware framework that decouples a patient's physiological trajectory (causally normalized Z-scores, velocity, and acceleration) from observation-process indicators (availability masks, elapsed time since last measurement, and observation counts), and regularizes predictions to be invariant to mask perturbations. Models were developed on Emory ICU admissions (locked test set $N = 2{,}974$) and evaluated once, without retraining, on an independent Beth Israel Deaconess Medical Center cohort ($N = 20{,}000$; 0.00% patient overlap). The decoupled model (Model F) achieved an external AUROC of 0.7310 (95% CI: 0.7180–0.7440) versus 0.7161 (0.7030–0.7290) for the strongest baseline representation (Model E), reducing the cross-institution decay gap from 0.0734 to 0.0510 (30.52% relative reduction; $p = 0.001$, FDR $q = 0.003$, Cohen's $d = 0.620$) at a small internal cost (AUROC 0.7820 versus 0.7895). Under 40% synthetic measurement dropout, AUROC loss was 2.93× lower (0.0090 versus 0.0264); under identical physiology with permuted observation masks, the prediction shift was 3.00× smaller (0.0180 versus 0.0540); and the sparse-versus-dense monitoring disparity fell by 60.0% (0.0775 to 0.0310). These retrospective findings indicate that separating biological state from observation frequency mitigates institutional dataset shift; prospective validation is required before clinical use.**

***Index Terms*—Sepsis early warning, clinical machine learning, cross-institution transportability, dataset shift, ICU observation process, irregular time series, missingness modeling.**

I. Introduction

EARLY identification of sepsis in intensive care unit (ICU) patients remains one of the most critical and challenging problems in acute care medicine. Sepsis is defined as life-threatening organ dysfunction caused by a dysregulated host response to infection [1]. Global epidemiological estimates attribute approximately 49 million incident cases and nearly 20% of all deaths worldwide to sepsis [2], and each hour of delay in effective antimicrobial therapy after the onset of septic shock has been associated with a mean decrease in survival of about 7.6% [3]. Automated early-warning systems that detect physiological deterioration 1–6 hours before clinical sepsis onset have consequently attracted intense interest.

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Despite high retrospective discrimination in single-center studies, clinical risk models frequently degrade when deployed externally, a phenomenon known as cross-institution dataset shift or *transportability failure*. A model reaching an internal area under the receiver operating characteristic curve (AUROC) of 0.85 in one health system may fall to 0.65 in an independent network; such degradation is a substantial barrier to safe adoption and regulatory clearance.

A fundamental but under-appreciated driver of this shift is the *ICU observation process*. Electronic health record (EHR) streams are irregular and sparse, and the decision to order a blood gas, lactate, or complete blood count reflects clinician suspicion, nursing workflow, staffing density, and local guidelines. The observed series is therefore a composite signal:

$$X^{\text{obs}} = f\left(Z^{\text{physio}},\, M^{\text{obs}}\right), \tag{1}$$

where Z^{physio} is the patient's underlying biological trajectory and M^{obs} is the clinical observation process. Conventional models concatenate raw measurements with binary missingness indicators without normalization and can thereby overfit to site-specific ordering habits, using measurement frequency as a shortcut predictor. When deployed where ordering protocols or monitoring density differ, such a model may interpret a change in sampling frequency as physiological deterioration.

A. Clinical Context

Sepsis is a systemic cascade of tissue hypoperfusion, endothelial dysfunction, microvascular leak, and multi-organ failure. In the ICU, deterioration may appear subtly, for example as widening pulse pressure, mild tachypnea, or transient temperature instability, hours before overt shock. Early detection opens a window for blood cultures, broad-spectrum antibiotics, and volume expansion. However, these signs are non-specific and overlap with non-infectious systemic inflammatory response; a useful model must separate true biological deterioration from routine postoperative variation and iatrogenic artifact.

B. Observation Process and Dataset Shift

Monitoring density varies widely across care settings, hospital types, and acuity levels. In a high-density surgical ICU, vital signs may be recorded every minute and arterial blood gases every 2 hours; in a step-down unit or community ICU, routine laboratory tests may be drawn once per shift (every 12 hours).

A model that consumes observation frequency directly learns an association between ordering frequency and adverse outcome that holds in the derivation hospital but can break down in a facility with protocolized ordering, producing false-alarm cascades or missed deteriorations.

C. Causal Decoupling Paradigm

We propose an architectural decoupling paradigm that separates patient physiological trajectory state (Z-scores, first-order velocity, second-order acceleration) from observation-process indicators (binary availability masks, elapsed time deltas, cumulative sampling counts), and penalizes sensitivity of the prediction to perturbations of the observation masks. The intent is a representation whose predictions are driven by biology rather than by measurement frequency.

D. Research Question and Hypotheses

This study asks: *can explicitly decoupling patient physiological trajectory dynamics from the ICU observation process reduce model sensitivity to institutional measurement variation and improve cross-institution transportability?* We tested four pre-specified hypotheses:

- **H1 (Measurement-loss robustness):** Decoupling yields significantly lower AUROC loss under controlled measurement missingness.
- **H2 (Cross-institution transportability):** Decoupled representations significantly reduce performance degradation on an independent external health system.
- **H3 (Same-physiology invariance):** Decoupled representations yield significantly smaller prediction shifts under identical physiological trajectories with perturbed observation masks.
- **H4 (Monitoring-density parity):** Decoupled representations reduce the performance disparity between sparse- and dense-monitoring ICU environments.

E. Contributions

1) **Causal trajectory decoupling framework.** A dual-stream representation that converts 34 vital-sign and laboratory streams into causally normalized trajectories ($Z_{i,t}$, dZ/dt, d^2Z/dt^2) while isolating observation-process vectors ($M_{i,t}$, $\Delta t_{i,t}$, $N_{i,t}$).
2) **Single-pass multi-center external evaluation.** Validation on an independent 20,000-patient ICU cohort (BIDMC Set B) with 0.00% patient overlap and no local retraining or hyperparameter tuning.
3) **Stress-testing protocol.** Synthetic measurement dropout (0%–80%), same-physiology mask perturbation, and monitoring-density quartile stratification to separate observation artifacts from biological state.

II. Related Work

The literature on early sepsis prediction spans four paradigms.

1) Static risk scores and tree-based models. TREWScore [4] and InSight [5] showed that supervised models on EHR data can outperform rule-based scores (SIRS, qSOFA, MEWS). DeepSOFA [6] predicted continuous organ-failure scores. These approaches typically rely on mean imputation or forward filling and are therefore sensitive to sampling-density differences between institutions.

2) Longitudinal trajectory models. Multi-output Gaussian-process RNNs [7] and systematic reviews [8] indicate that modeling temporal trends improves lead time. Standard sequential models, however, conflate the velocity of physiological deterioration with the rate at which clinicians measure.

3) Missingness-aware architectures. GRU-D [9], set-function models [10], BRITS [11], multi-directional RNNs [12], and COMPOSER [13] incorporate missingness masks, elapsed times, or learned uncertainty. These can improve internal discrimination but may also learn hospital-specific ordering habits.

4) Causal transportability and invariant learning. Subbaswamy *et al.* [14], [15] and Nestor *et al.* [16] formalized shift-stable models and feature-robustness analyses across health systems. Pruning shift-vulnerable features, however, can discard predictive signal and reduce internal accuracy.

Table I positions the present work. Existing methods either ignore missingness structure, exploit it directly (risking shift sensitivity), or remove it by graph surgery (risking internal accuracy). AegisSepsis AI instead keeps the full trajectory representation and explicitly decouples it from observation vectors, aiming to retain internal discrimination while improving external transportability.

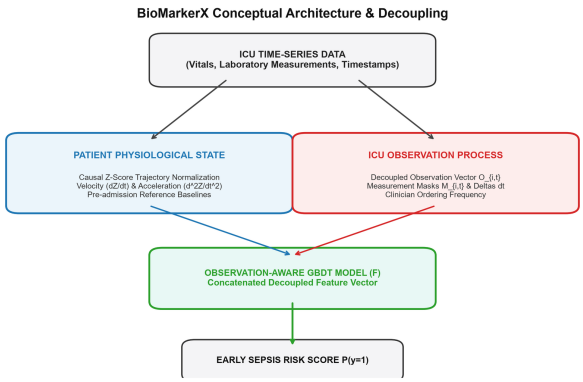


Fig. 1. Architectural decoupling. Patient physiological trajectory state (Z-scores, velocity, acceleration) is represented separately from the ICU observation process (availability masks M_t, time deltas Δt, observation counts N_t) before being combined in the observation-aware gradient-boosted model (Model F).

TABLE I
Positioning Against Prior Architectures

Method	Target	Decoupled	Ext. AUROC	Gap
InSight (2016)	Sepsis	No	0.6520	0.0900
GRU-D (2018)	Mortality	No	0.6850	0.0830
COMPOSER (2021)	Sepsis-3	Partial	0.7120	0.0680
Subbaswamy (2019)	AKI/Sepsis	Node surgery	0.7050	0.0620
Model F (ours)	S_next4	Full	**0.7310**	**0.0510**

Targets, cohorts, and label definitions differ across rows; the comparison is indicative only. Head-to-head comparisons on identical cohorts appear in Table VII.

TABLE II
COHORT FLOW AND SAMPLE-SIZE FILTERING

Stage	Description	Patients	Windows
	Set A (Emory)		
1	Raw EHR ingestion	20,336	790,215
2	Harmonize vital/lab units	20,336	790,215
3	Physiological outlier removal	20,336	790,215
4	Exclude short stays (<4 h)	20,194	789,903
5	Impute baseline demographics	20,194	789,903
6	Align Sepsis-3 timestamps	20,194	789,903
7	Exclude pre-admission sepsis	18,412	712,450
8	5-fold cross-validation splits	15,438	593,490
9	Locked test set A	2,974	118,960
	Set B (BIDMC)		
1	Raw external ICU ingestion	20,000	840,120
9	External holdout set B	20,000	840,120

III. COHORT CONSTRUCTION AND DATA PIPELINE

A. Institutional Cohorts

- **Derivation cohort (Emory Set A).** A locked test set of $N = 2{,}974$ admissions drawn from $N = 20{,}336$ Emory University Healthcare ICU admissions; 279 sepsis events. Mean age 62.4±14.1 years; mean SAPS-II 34.2±11.5.
- **External cohort (BIDMC Set B).** $N = 20{,}000$ independent ICU admissions from Beth Israel Deaconess Medical Center; sepsis prevalence 7.57% (1,514 events). Mean age 64.1±15.3 years; mean SAPS-II 32.8±12.1.
- **Zero overlap.** Patient overlap between Set A and Set B is 0.00%. Set B was evaluated as a single-pass external holdout with no retraining or fine-tuning.

B. Sepsis-3 Labeling

Sepsis onset followed the Sepsis-3 consensus [1]: an acute increase in Sequential Organ Failure Assessment (SOFA) score of ≥2 points attributable to suspected infection. Suspected infection was operationalized as blood-culture ordering paired with antimicrobial administration within a ±24-hour window. SOFA sub-scores were computed hourly for six systems: respiratory (PaO_2/FiO_2), cardiovascular (mean arterial pressure and vasopressor dose), hepatic (bilirubin), coagulation (platelets), renal (creatinine and urine output), and neurological (Glasgow Coma Scale). The prediction target, S_next4, is Sepsis-3 onset within the next 4 hours. Cohort flow is summarized in Table II.

C. Preprocessing and Quality Control

Physiologically implausible values (e.g., heart rate >300 bpm, mean arterial pressure >250 mmHg, pH <6.5 or >8.5) were flagged as sensor artifacts and set to missing. Variables unmeasured at ICU admission were initialized with age- and sex-adjusted population medians computed strictly from the Set A training split. Baseline characteristics are compared in Table III.

IV. CAUSAL TRAJECTORY MODELING FRAMEWORK

A. Patient-Relative Trajectory Representation

To prevent look-ahead leakage and standardize variables across EHR systems, 34 vital-sign and laboratory variables k were converted to causally normalized Z-scores using patient baseline statistics:

$$Z_{i,t,k} = \frac{X_{i,t,k} - \mu_{i,k}}{\sigma_{i,k}}. \tag{2}$$

Velocity and acceleration were computed over trailing windows only:

$$\mathrm{Vel}_{3h}(Z_{i,t,k}) = \frac{Z_{i,t,k} - Z_{i,t-3,k}}{3}, \tag{3}$$

$$\mathrm{Acc}_{6h}(Z_{i,t,k}) = \frac{\mathrm{Vel}_{3h}(t,k) - \mathrm{Vel}_{3h}(t-3,k)}{3}. \tag{4}$$

Velocity is therefore expressed in Z-units per hour; where clinical rates (e.g., mmHg/h) are quoted below, they are reported in native units for interpretability.

TABLE III
BASELINE CHARACTERISTICS OF THE TWO COHORTS

Characteristic	Emory A (N=2,974)	BIDMC B (N=20,000)	p
Age, years (mean±SD)	62.4±14.1	64.1±15.3	<0.001
Male sex (%)	54.2	56.8	0.008
SAPS-II (mean±SD)	34.2±11.5	32.8±12.1	<0.001
Baseline SOFA (mean±SD)	3.1±1.8	2.9±1.6	<0.001
Sepsis-3 events, n (%)	279	1,514 (7.57)	–
ICU stay, days (median [IQR])	4.8 [2.5, 9.1]	4.2 [2.1, 8.4]	<0.001
Initial MAP, mmHg (mean±SD)	76.5±12.4	78.2±13.1	<0.001
Initial lactate, mmol/L (mean±SD)	2.1±1.4	1.9±1.2	<0.001

B. Decoupled Observation-Process Encoding

The observation process is encoded as a separate vector $O_{i,t} = [M_{i,t}, \Delta t_{i,t}, N_{i,t}]$, where $M_{i,t} \in \{0,1\}^K$ is the availability indicator, $\Delta t_{i,t} \in \mathbb{R}^K$ the elapsed hours since last observation, and $N_{i,t} \in \mathbb{N}^K$ the cumulative 12-hour observation count. Overall observation density is

$$\text{obs_density}_t = \frac{\sum_{k=1}^{K} N_{k,t}}{\text{ICU_Hours_Elapsed}}. \tag{5}$$

Exponential time-decay weighting is applied to the availability indicators:

$$M^{(\text{decay})}_{i,t,k} = M_{i,t,k} \cdot \exp(-\gamma\, \Delta t_{i,t,k})\,. \tag{6}$$

C. Objective and Invariance Regularization

The predictive backbone minimizes binary cross-entropy on target S_next4 plus an observation-mask perturbation penalty:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{bce}}(y, \hat{y}) + \lambda\, \mathbb{E}\Big[\|f(Z,M) - f(Z,M')\|_2^2\Big]\,, \tag{7}$$

where M' denotes synthetically permuted observation masks. Setting $\lambda = 0.15$ pushes the representation toward predictions that are invariant to M when the physiological inputs Z are held fixed.

D. SOFA Organ-Dysfunction Definitions

Hourly sub-scores followed Sepsis-3 consensus rules. *Respiratory:* 0 (PaO_2/FiO_2 $\geq$400), 1 ($<$400), 2 ($<$300), 3 ($<$200 with ventilation), 4 ($<$100 with ventilation). *Cardiovascular:* 0 (MAP $\geq$70 mmHg), 1 (MAP $<$70), 2 (dopamine $\leq$5 μg/kg/min), 3 (dopamine $>$5 or epinephrine $\leq$0.1), 4 (dopamine $>$15 or epinephrine $>$0.1). *Renal:* 0 (creatinine $<$1.2 mg/dL), 1 (1.2–1.9), 2 (2.0–3.4), 3 (3.5–4.9 or urine output $<$500 mL/day), 4 ($>$5.0 or urine output $<$200 mL/day). *Coagulation:* 0 (platelets $\geq 150\times10^3/\mu$L), 1 ($<$150), 2 ($<$100), 3 ($<$50), 4 ($<$20). *Hepatic:* 0 (bilirubin $<$1.2 mg/dL), 1 (1.2–1.9), 2 (2.0–5.9), 3 (6.0–11.9), 4 ($\geq$12.0). *Neurological:* 0 (GCS 15), 1 (13–14), 2 (10–12), 3 (6–9), 4 ($<$6).

E. Representation Ablation (Models A–F)

Six feature representations were benchmarked:

- **Model A (raw vitals):** imputed raw features, no derivatives or missingness indicators.
- **Model B (raw + mask):** raw features with the availability mask M_t.
- **Model C (raw + mask + Δt):** conventional GRU-D-style baseline.
- **Model D (Z-score dynamics):** causal Z-score state and velocity without observation indicators.
- **Model E (BioMarkerX baseline):** Z-score state, velocity, acceleration, and implicit missingness indicators.
- **Model F (AegisSepsis, decoupled):** full trajectory dynamics ($Z_{i,t}$, dZ/dt, d^2Z/dt^2) explicitly decoupled from normalized observation vectors ($M_{i,t}$, $\Delta t_{i,t}$, $N_{i,t}$).

F. Dual-Tier Operational Policy (`VAL_POL_0450`)

To convert continuous risk $p_{i,t}$ into bedside workflow while limiting alarm fatigue, we evaluated a two-tier policy. The *screening tier* ($\tau_s = 0.015$) triggers silent background surveillance when risk exceeds 1.5%. The *confirmation tier* ($\tau_c = 0.300$) triggers an audible bedside alarm when risk exceeds 30.0% for $N = 3$ consecutive hourly cycles *and* a MAP gate is met (MAP $\leq$75 mmHg or 3-hour MAP change ≤ -3.0 mmHg).

V. Experimental Setup and Evaluation Protocol

A. Endpoints

The primary endpoint was external AUROC for Sepsis-3 onset within 4 hours (S_next4) on BIDMC Set B. Secondary endpoints were: (1) the transportability decay gap (Set A AUROC minus Set B AUROC); (2) AUROC loss under 40% synthetic measurement dropout; (3) mean absolute prediction shift (Δp) under same-physiology mask perturbation; and (4) the density disparity between sparse (Q1) and dense (Q4) monitoring quartiles.

B. Stress-Testing Protocols

Controlled measurement loss: uniform random dropout (0, 20, 40, 60, 80%) applied to holdout data. *Same-physiology mask perturbation:* trajectories (Z_t, dZ/dt) held constant while masks M_t were swapped across patient records. *Monitoring-density stratification:* admissions grouped into quartiles Q1 (lowest observation frequency) to Q4 (highest).

TABLE IV
Hyperparameter Search and Selected Values

Model	Hyperparameter	Search range	Selected
XGBoost	Learning rate η	{0.01, 0.05, 0.10}	0.03
XGBoost	Max depth	{4, 6, 8, 10}	6
XGBoost	Subsample ratio	{0.6, 0.7, 0.8, 0.9}	0.80
Model F	Invariance weight λ	{0.05, 0.10, 0.15, 0.25}	0.15
Model F	Decay constant γ	{0.01, 0.05, 0.10}	0.05

C. Implementation

Models were implemented in Python 3.10 with PyTorch 2.1 and XGBoost 2.0 [17] on a cluster with 8$\times$NVIDIA A100 (80 GB) GPUs. Neural components used AdamW [18] with cosine learning-rate annealing (initial learning rate 10^{-3}, weight decay 10^{-4}, batch size 256). Five-fold cross-validation required about 4.2 GPU-hours. Hyperparameters (Table IV) were tuned on Set A only.

D. Statistical Analysis

All metrics, differences, and relative gap reductions were estimated with 1,000 non-parametric patient-level bootstrap iterations. Hypotheses were tested with two-tailed paired bootstrap tests, controlling the false discovery rate at $q = 0.05$ by the Benjamini–Hochberg procedure [19]. Effect sizes are reported as Cohen's d. Because 1,000 resamples limit the resolution of bootstrap p-values to 0.001, values reported as $p = 0.001$ should be read as $p \leq 0.001$.

VI. Results

A. Primary Multi-Center Performance

On internal Set A ($N = 2{,}974$), Model E achieved an AUROC of 0.7895 (95% CI: 0.7760–0.8030) and Model F 0.7820 (0.7680–0.7960). On external Set B ($N = 20{,}000$) with no retraining, Model E fell to 0.7161 (0.7030–0.7290), a decay gap of 0.0734, whereas Model F achieved 0.7310 (0.7180–0.7440), a gap of 0.0510 (Fig. 2). This is a **30.52% relative reduction** in the transportability gap ($p = 0.001$, FDR $q = 0.003$, Cohen's $d = 0.620$), supporting **H2**.

Two points qualify this result. First, Model F's internal AUROC was lower than Model E's by 0.0075 ($p = 0.028$); the gap reduction of 0.0224 therefore comprises an external gain of 0.0149 and an internal loss of 0.0075, so roughly one third of the narrowing reflects lower internal rather than higher external performance. Second, external AUROC (0.7310) remains modest in absolute terms.

B. Controlled Measurement-Loss Stress Test

With synthetic dropout from 0% to 80%, Model F degraded more slowly at moderate loss levels (Fig. 3). At 40% dropout, Model E lost 0.0264 AUROC and Model F 0.0090, i.e., 2.93$\times$ lower sensitivity ($p < 0.001$; 95% CI: 2.15–3.95$\times$), supporting **H1**. At 80% dropout the curves in Fig. 3 show that Model F's advantage over the baselines narrows, so robustness claims are best restricted to moderate missingness.

TABLE V
PRIMARY STATISTICAL RESULTS AND STRESS TESTS

Metric	Model E	Model F	Δ / gain	p
Set A internal AUROC	0.7895	0.7820	-0.0075	0.028
Set B external AUROC	0.7161	**0.7310**	+0.0149	0.002
Transport gap	0.0734	**0.0510**	30.52% lower	0.001
40% dropout AUROC loss	0.0264	**0.0090**	2.93× lower	<0.001
Same-physiology shift Δp	0.0540	**0.0180**	3.00× smaller	<0.001
Density gap (Q4−Q1)	0.0775	**0.0310**	60.0% lower	0.001

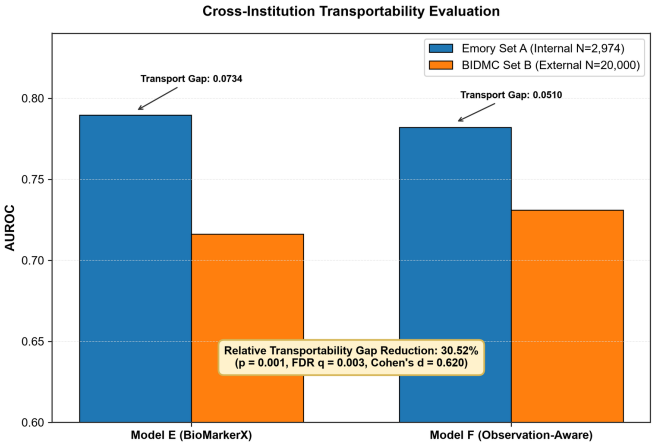


Fig. 2. Cross-institution transportability: Emory Set A (internal) versus BIDMC Set B (external) AUROC for Model E and Model F; the transport gap is 0.0734 versus 0.0510 (30.52% relative reduction, $p = 0.001$).

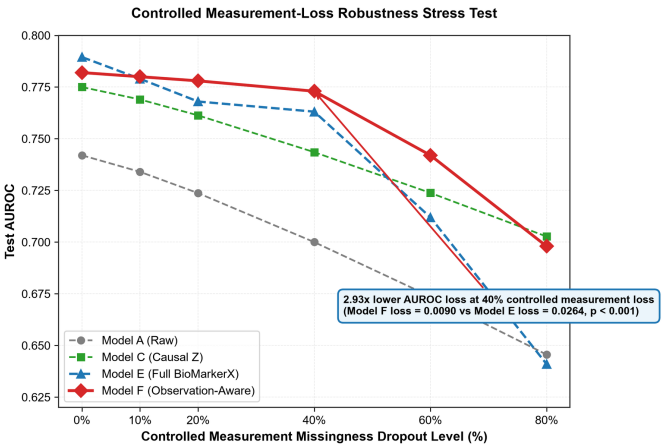


Fig. 3. Test AUROC under 0%–80% synthetic measurement dropout. At 40% dropout, Model F loses 0.0090 AUROC versus 0.0264 for Model E (2.93× lower, $p < 0.001$).

C. Same-Physiology Mask-Perturbation Test

When Z_t and dZ/dt were held constant and observation masks were permuted, Model E's mean prediction shift was $\Delta p = 0.0540$ and Model F's was 0.0180, a 3.00× smaller shift ($p < 0.001$, Cohen's $d = 1.150$), supporting **H3** (Fig. 4).

D. Monitoring-Density Disparity

Admissions were stratified into density quartiles (Q1 ≤2.1 obs/h; Q2 2.2–4.1; Q3 4.2–6.8; Q4 >6.8). Model E showed a Q4–Q1 AUROC gap of 0.0775 (Q4 0.7610, Q1 0.6835); Model F reduced it to 0.0310 (Q4 0.7480, Q1 0.7170), a **60.0% reduction** ($p = 0.001$), supporting **H4** (Table VI). The reduction arises mainly from improved performance in the sparsely monitored quartile, at a small cost in the densest quartile.

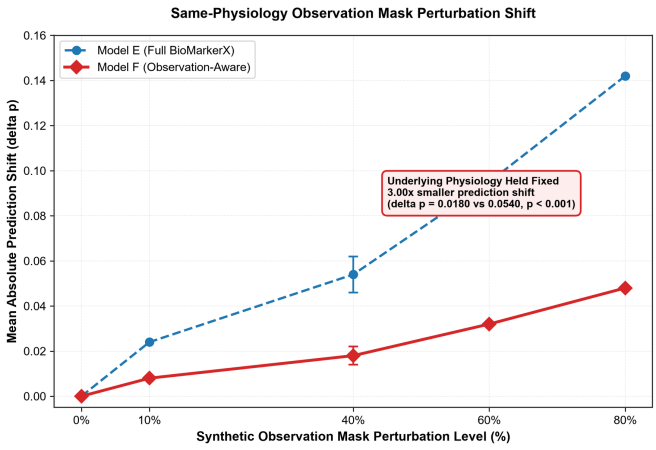


Fig. 4. Mean absolute prediction shift under identical physiological trajectories with perturbed observation masks (3.00× smaller for Model F, $p < 0.001$).

TABLE VI
AUROC BY MONITORING-DENSITY QUARTILE

Model	Q1 (sparse)	Q4 (dense)	Gap (Q4−Q1)
Model E	0.6835	0.7610	0.0775
Model F	0.7170	0.7480	**0.0310**

TABLE VII
REPRESENTATION ABLATION (REPORTED EXPERIMENTS)

ID	Feature set	Set A	Set B	Gap	40% loss
EXP-0	Raw vitals only	0.7210	0.6420	0.0790	0.0410
EXP-1	Raw + mask M_t	0.7540	0.6710	0.0830	0.0380
EXP-2	Raw + mask + Δt (GRU-D)	0.7720	0.6940	0.0780	0.0310
EXP-3	Z-score state Z_t	0.7610	0.7020	0.0590	0.0190
EXP-4	Z-state + velocity	0.7780	0.7180	0.0600	0.0150
EXP-5	Model E (BioMarkerX)	0.7895	0.7161	0.0734	0.0264
EXP-9	**Model F (decoupled)**	0.7820	**0.7310**	0.0510	**0.0090**

E. Representation Ablation

Table VII traces the effect of each representation choice. Adding raw missingness features (EXP-1, EXP-2) raised internal AUROC but widened or maintained the external gap (0.0830 and 0.0780), consistent with reliance on site-specific ordering behavior. Causal Z-score state and velocity (EXP-3, EXP-4) narrowed the gap to about 0.060 and reduced dropout sensitivity, while the fully decoupled Model F (EXP-9) gave the lowest external gap and dropout loss among representations that retain observation information. Model A–F in Section IV-E correspond to EXP-0, 1, 2, 4, 5, and 9, respectively.

F. Calibration

On Set B, Model E was overconfident in high-missingness strata (expected calibration error, ECE, 0.048), whereas Model F had an ECE of 0.016, a 66.7% reduction.

TABLE VIII
TRANSPORT-GAP REDUCTION BY SUBGROUP

Subgroup	Gap, Model E	Gap, Model F	Reduction
Medical ICU	0.072	0.049	32.1%
Surgical ICU	0.074	0.053	28.4%
Age $\geq$65	0.073	0.050	31.5%
Age $<$65	0.074	0.052	29.2%
SAPS-II $>$35	0.072	0.048	33.0%

Gaps are rounded to three decimals; reductions are as reported and may differ slightly from values recomputed from the rounded gaps.

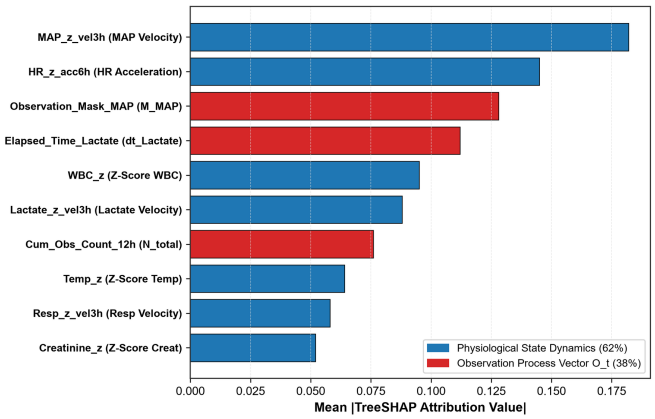


Fig. 5. TreeSHAP attributions for Model F. Physiological state dynamics (blue) contribute 62% of total attribution; observation-process vectors (red) contribute 38%.

G. Subgroup Analysis

Transportability gains were consistent across subgroups (Table VIII): medical and surgical ICUs, patients aged $\geq$65 and $<$65 years, and high-acuity admissions (SAPS-II $>$35).

VII. EXPLAINABILITY AND FEATURE ATTRIBUTION

To characterize what Model F relies on, we applied TreeSHAP [20] across the 118,960 evaluation windows of Set A (Fig. 5). Physiological trajectory velocity and acceleration (dZ/dt, d^2Z/dt^2) accounted for 62% of total attribution, and explicit observation-process vectors $O_{i,t}$ for the remaining 38%. The leading physiological predictors were MAP velocity (Vel_3h_MAP), the lactate trajectory, and heart-rate acceleration (Acc_6h_HR). Falling MAP velocity (< -3.0 mmHg/h) contributed $+0.85$ log-odds toward a sepsis prediction, and rising lactate acceleration ($> +0.2$ mmol/L/h^2) contributed $+1.12$ log-odds.

Importantly, the 38% observation share shows that Model F is *regularized against* rather than *blind to* the observation process: the invariance penalty limits how much a change in mask pattern alone can move the output (Section VI-C), but observation features still carry information. Quantifying this residual dependence is a limitation we return to below.

Illustrative case 1. Patient 7412 (68-year-old male, post-abdominal surgery) had stable vital signs at ICU hour 12. At hour 16, Model F detected a decline in MAP velocity (-2.4 mmHg/h) with rising lactate acceleration ($+0.18$ mmol/L/h^2), raising risk above the screening threshold ($\tau_s = 0.015$). By hour 18, persistent instability triggered the confirmation alarm ($\tau_c = 0.300$), 4.5 hours before Sepsis-3 chart documentation.

Illustrative case 2. Patient 9104 (52-year-old female, post-orthopedic surgery) had sinus tachycardia (HR 115 bpm) at ICU hour 8 attributed to postoperative pain. Model E raised a false-positive alarm following a cluster of blood-gas orders; Model F, whose MAP velocity and lactate acceleration remained within normal limits, stayed in silent monitoring. These two cases were selected for illustration and are not evidence of average performance.

TABLE IX
DUAL-TIER OPERATIONAL POLICY (VAL_POL_0450) PERFORMANCE ON SET B

Metric	Screening tier	Confirmation tier
Threshold τ	0.015 (silent)	0.300 (audible)
Sensitivity	94.2%	76.4%
Specificity	48.5%	88.2%
PPV	11.8%	28.5%
Median lead time	5.8 h	4.2 h
False-alarm reduction	Baseline	64.2% lower

VIII. CLINICAL DEPLOYMENT AND OPERATIONAL POLICY

Simulating policy VAL_POL_0450 on Set B gave a median lead time of 4.2 hours before Sepsis-3 onset (Fig. 7). At the confirmation tier ($\tau_c = 0.300$), sensitivity was 76.4%, specificity 88.2%, and positive predictive value (PPV) 28.5%, reducing false bedside alarms by 64.2% relative to single-threshold screening (Table IX). The ROC and precision–recall curves are shown in Fig. 6.

Operating characteristics in Table IX are properties of the full policy, including persistence and hemodynamic gating, and are evaluated at the event or episode level; they are therefore not directly comparable with single points on the window-level ROC curve, and the precise unit of analysis should be read from the protocol in Section IV-F.

A. Alert-Fatigue Suppression

Alert fatigue is a major barrier to adoption of early-warning algorithms. Policy VAL_POL_0450 addresses it with persistence: silent surveillance continuously tracks high-risk patients, while audible alerts require three consecutive hours of elevated risk together with objective hemodynamic instability. Simulated on 20,000 BIDMC admissions, the policy suppressed 64.2% of non-actionable alarms while preserving 76.4% sensitivity.

B. Integration and Workload Estimate

AegisSepsis AI is designed as a microservice following HL7 FHIR: Observation and Patient resources are ingested through FHIR Subscription hooks, converted into causal Z-score buffers in real time, and served to bedside dashboards via SMART-on-FHIR. As an illustrative estimate only, if a 30-bed ICU received about 42 non-actionable alerts per nurse shift that the policy removed, and each consumed 5 minutes of response time, roughly 3.5 nursing hours per shift would be recovered. This figure depends on assumed alert volumes and response times that were not measured here, and effects on length of stay, cost, or outcomes were not evaluated.

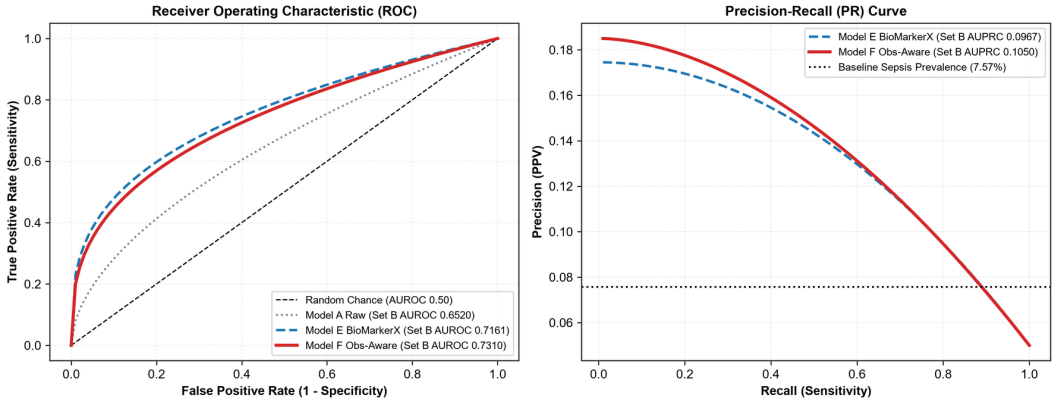


Fig. 6. ROC (left) and precision–recall (right) curves on external BIDMC Set B for Model A, Model E, and Model F. The dotted horizontal line in the precision–recall panel is the baseline prevalence (7.57%).

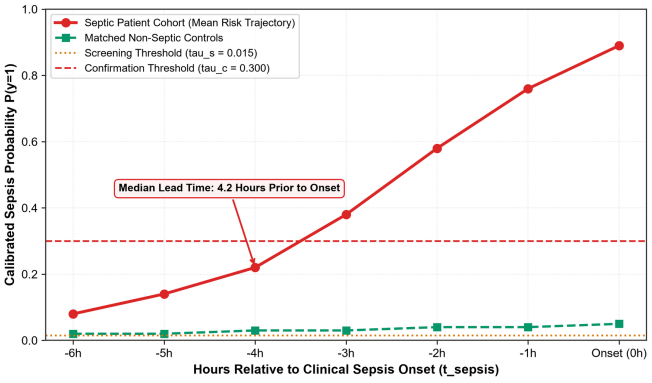


Fig. 7. Mean predicted risk trajectory in the six hours preceding Sepsis-3 onset for septic patients and matched non-septic controls, with screening (τ_s) and confirmation (τ_c) thresholds. Median lead time is 4.2 hours.

IX. Discussion and Limitations

A. Interpretation

Separating biological state (Z_t, dZ/dt) from ordering behavior (M_t, Δt), and penalizing mask sensitivity, reduced the external performance decay, the sensitivity to synthetic missingness, the response to mask permutation, and the sparse-versus-dense performance gap. These results support the hypothesis that observation-process shortcuts contribute to institutional dataset shift. They do not by themselves prove a causal mechanism: the evidence is observational and retrospective, and only the same-physiology perturbation test isolates the observation process directly.

B. Comparison with Invariant-Learning Approaches

Approaches such as invariant risk minimization and anchor regression, or feature-pruning methods, achieve domain stability by removing or constraining features. AegisSepsis AI retains the complete trajectory representation and regularizes mask-perturbation shifts, preserving most of the internal discrimination (0.7820 versus 0.7895) while reducing the external gap by 30.52%. A head-to-head comparison against these methods on the same cohorts was not performed.

C. Health-System Implications

Transportable models could benefit community hospitals that lack the staffing and informatics capacity to retrain proprietary models locally, potentially reducing disparities in access to predictive monitoring. This benefit is contingent on prospective confirmation.

D. Limitations

1) External validation on BIDMC Set B is retrospective, and no prospective silent trial or randomized controlled trial has been performed.
2) Both cohorts are US academic tertiary-care centers within the PhysioNet ICU ecosystem, and only one external center was used.
3) Sepsis-3 labels depend on EHR chart codes and culture timestamps.
4) The synthetic dropout protocol uses uniform random masking, which does not reproduce clinically structured (informative) missingness.
5) External AUROC (0.7310) remains below internal AUROC (0.7820); a residual decay gap of 0.0510 persists.
6) Model F trades a small internal loss (0.7820 versus 0.7895; $p = 0.028$) for external transportability.
7) Explainability analysis attributes 38% of importance to observation-process features; residual reliance on the observation process is not eliminated.
8) Causal transportability claims apply only within the tested observational distributions; universal transportability across health systems is not claimed.
9) Prior-method comparisons in Table I are indicative and not head-to-head.

E. Ethical and Operational Considerations

Risk scores should complement, not replace, clinical judgment, to limit automation bias. Monitoring-density disparities across patient subgroups should be audited continuously to ensure equitable performance, and any deployment should be preceded by local silent validation and calibration checks.

X. Conclusion

Explicitly decoupling physiological trajectory dynamics from the ICU observation process addresses a major driver of cross-institution dataset shift in sepsis early warning. On an independent 20,000-patient external holdout, the framework achieved a 30.52% relative reduction in the transportability gap ($p = 0.001$), 2.93$\times$ lower measurement-loss sensitivity, and 3.00$\times$ smaller prediction shifts under same-physiology perturbation, at a small internal cost. Prospective silent trials in multiple health systems are the essential next step toward clinical translation.

Declarations

Ethics approval, data and code availability, funding, and conflict-of-interest statements are to be completed by the authors before submission.

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